PRODUCT LINE PROFITABILITY & MARGIN PERFORMANCE ANALYSIS

A Business Analytics Project Report

Case Study: Nassau Candy Distributor

Prepared using Python, Pandas, and Streamlit

Project report for academic / professional submission

Abstract

This project analyzes product-level and division-level profitability for Nassau Candy Distributor. It examines which products contribute most to sales and gross profit, whether high-sales products generate healthy margins, and where margin risk may require management attention. Python and Pandas support data preparation and calculations, while Streamlit presents results in an interactive dashboard. Techniques include gross-margin calculation, product ranking, division comparison, margin-risk screening, and Pareto concentration analysis. Reported aggregate results are total sales of ■141,783.63, gross profit of ■93,442.80, and overall gross margin of approximately 65.91%.

1. Introduction

Product profitability analysis helps an organization understand how revenue translates into gross profit across products and divisions. A high-sales product may not necessarily have a strong margin, while a lower-sales product may generate a comparatively higher margin. Reviewing these measures together supports pricing, product-mix, promotion, and inventory decisions.

This project applies business analytics to product sales data associated with Nassau Candy Distributor. It provides an evidence-based view of product performance and communicates findings through an interactive Streamlit dashboard.

2. Problem Statement

A sales-only view can conceal low-margin products, profitability concentration, and differences between divisions. The project establishes a structured approach to evaluate sales, gross profit, gross margin, profit per unit, and contribution measures, highlighting products whose sales may not translate into proportionate gross profit.

3. Objectives

- Identify products with the highest and lowest gross-margin performance.
- Determine whether high-sales products also generate healthy margins.
- Compare sales, gross profit, and margin across divisions.
- Flag products below a selected margin threshold for review.
- Measure revenue and gross-profit concentration among leading products.
- Present findings in an interactive dashboard.

4. Scope

The project covers data preparation, exploratory analysis, metric calculation, product ranking, division comparisons, margin diagnostics, Pareto analysis, and dashboard presentation. Controls include date range, division, margin threshold, and product search. It does not establish causal explanations for low margins, and excludes State/Region analysis and a complete raw-data table.

5. Dataset Overview

The data contains transaction-level sales information and fields used to calculate profitability. Key fields include Product Name, Sales, Gross Profit, Units, Division, and Order Date (where available).

A transaction count of approximately 10,194 was referenced during development; verify the final count against the processed CSV before submission.

6. Tools and Technologies

- Python for programming and analysis.

- Pandas for cleaning, grouping, and aggregation.
- Plotly for interactive charts.
- Streamlit for the dashboard interface.
- CSV for processed-data exchange.
- Visual Studio Code / Jupyter Notebook for development.

7. Data Preparation and Validation

The preparation stage checks column names and data types, converts dates, reviews missing values and duplicate records, and validates numeric sales, profit, and units. Product labels should be standardized so spelling variants do not split a product into separate categories. Calculations should handle zero denominators and avoid invalid or infinite values.

8. Methodology

8.1 Exploratory Data Analysis

Exploratory analysis reviews dataset structure, product and division categories, numeric distributions, and ranges of sales and gross-profit values. Grouped summaries by product and division form the basis of the comparisons.

8.2 Profitability Metrics

Gross Margin (%) = (Gross Profit ÷ Sales) × 100
Profit per Unit = Gross Profit ÷ Units
Revenue Contribution (%) = Product Sales ÷ Total Sales × 100
Profit Contribution (%) = Product Gross Profit ÷ Total Gross Profit × 100

8.3 Product-Level Analysis

Records are grouped by product to calculate total sales, gross profit, margin, and contribution. Products are ranked to identify leaders and candidates for review. High-sales products are identified using a sales cutoff (upper quartile in the dashboard logic), then compared with the selected margin threshold.

8.4 Division-Level Analysis

Sales and gross profit are aggregated by division. Division margin is compared to distinguish business scale from the margin generated.

8.5 Margin-Risk Diagnostics

Products below a selected gross-margin threshold are flagged as candidates for further review. This is a screening indicator, not proof of unprofitability or of a specific cause. Additional analysis may consider price, discounts, cost inputs, and transaction mix.

8.6 Pareto Concentration Analysis

Products are ranked by sales or gross profit and cumulative contribution is calculated. The number of products needed to reach an 80% contribution level is determined from the first cumulative percentage at or above 80%.

9. Key Performance Indicators

KPI	Reported Value	Meaning
Total Sales	■141,783.63	Total sales value
Gross Profit	■93,442.80	Gross profit before operating expenses
Units	38,654	Total units represented

KPI	Reported Value	Meaning
Overall Gross Margin	65.91%	Gross profit as a share of sales

10. Results and Analysis

10.1 Division Performance

Division	Sales	Gross Profit	Average Margin
Chocolate	■131,692.90	■88,824.62	67.46%
Other	■9,663.25	■4,333.45	37.67%
Sugar	■427.48	■284.73	57.69%

Chocolate is the largest division by sales and gross profit and reports a 67.46% average margin. Other has lower sales and the lowest reported average margin at 37.67%, making it a useful area for review. Sugar has a reported margin of 57.69%, but its total sales are very small. Margin percentage and business scale should therefore be considered together.

10.2 Product Profitability

Reported examples of strong average margins include Wonka Bar - Nutty Crunch Surprise at approximately 71.35%, Wonka Bar - Scrumdiddlyumptious at approximately 69.44%, and Wonka Bar - Fudge Mallows at approximately 66.67%. These values should be reconciled with the final processed dataset.

High sales do not guarantee a high margin. A product may generate substantial gross profit through volume while still having a margin percentage that signals possible pricing, discount, or cost concerns.

10.3 High-Sales / Low-Margin Products

- Wonka Bar - Milk Chocolate
- Fizzy Lifting Drinks
- Kazookles
- Wonka Bar - Triple Dazzle Caramel
- Lickable Wallpaper
- Wonka Gum

Kazookles is a reported margin-risk example: sales of ■1,205.75, gross profit of ■92.75, and average gross margin of approximately 7.69%. This warrants investigation, but does not establish the cause; pricing and cost details should be examined.

10.4 Revenue and Profit Concentration

The top four products account for approximately 76.26% of revenue and 77.30% of gross profit. The top five account for approximately 92.88% of revenue and 95.06% of gross profit. Five of the fifteen products are reported to be needed to reach approximately 80% of both revenue and profit. This indicates a substantial dependence on a small product group.

11. Business Questions and Answers

Which products deliver the highest gross margin?

The product margin ranking identifies products with stronger gross-profit yield relative to sales. Reported examples include Wonka Bar - Nutty Crunch Surprise and Wonka Bar - Scrumdiddlyumptious. Product names should be

displayed with the corresponding margin values.

Are high-sales products actually profitable?

Not necessarily at the same margin level. The high-sales/low-margin examples listed above show that sales volume should be evaluated alongside margin. The dashboard's cutoff and margin threshold make the comparison adjustable.

How does profitability vary across divisions?

Chocolate leads in sales and gross profit and has the highest reported average margin. Other has the lowest reported average margin. Sugar has a higher margin than Other but a much smaller sales base.

Which products represent margin risk?

Products below the selected threshold are flagged for review. Kazookles is a notable reported example at approximately 7.69% average margin. A flag prioritizes investigation; it does not establish a cause.

How concentrated are revenue and profit?

The top five products represent approximately 92.88% of revenue and 95.06% of gross profit in the reported results, showing substantial concentration.

12. Streamlit Dashboard

The dashboard presents the analysis through a business-friendly interface and contains these modules:

- **Product Profitability Overview:** product margin leaderboard and profit-contribution charts.
- **Division Performance:** sales, gross profit, and margin comparisons.
- **Cost vs Margin Diagnostics:** diagnostic views and margin-risk indicators, where fields are available.
- **Profit Concentration:** Pareto charts and dependency indicators.
- **Business Questions and Answers:** filter-responsive answers.

Controls include date range, division, margin threshold, and product search. KPI cards and charts should be clearly labeled, with values and units readable.

13. Key Findings

- Reported overall gross margin is approximately 65.91%.
- Chocolate dominates sales and gross profit.
- Other has the lowest reported division average margin.
- Several high-sales products fall below the selected margin threshold.
- Kazookles is an example of a low-margin product requiring review.
- Revenue and gross profit are concentrated among a small set of products.

14. Recommendations

1. Review costs, discounts, pricing, and transaction mix for products below the margin threshold.
2. Evaluate whether pricing or cost improvements can lift margins on high-sales/low-margin products without materially reducing demand.
3. Examine product-level drivers of the Other division's lower margin.
4. Monitor availability and profitability of leading products due to their large contribution.
5. Use a consistent margin threshold for recurring monitoring.

6. Reconcile dashboard totals with source data before making business decisions.

15. Limitations

- Detailed cost components, discounts, and operational drivers may not be available.
- A low margin does not by itself identify its cause.
- Results depend on data quality, product-label consistency, and margin aggregation method.
- All reported figures should be validated against the final processed CSV.
- The analysis describes patterns and does not prove causal relationships.

16. Future Enhancements

- Add monthly and seasonal profitability trends.
- Measure margin volatility by product.
- Include discount and cost fields for deeper diagnostics.
- Forecast sales and gross-profit trends if sufficient history is available.
- Add alerts for products below target margin.
- Provide exportable filtered reports.

17. Conclusion

This project demonstrates how profitability analytics can support decisions beyond a simple sales ranking. Combining sales, gross profit, margin, division comparison, and Pareto concentration identifies leading products, margin-review candidates, and the extent of dependence on a small product group. The Streamlit dashboard makes these findings accessible through interactive filters and visual summaries. The results can support pricing, product-mix, and monitoring discussions, provided the figures are reconciled with the final dataset and flagged products are investigated before action.

Appendix A: Project Workflow

1. Source data: load and inspect transaction data.
2. Preparation: clean labels, validate values, parse dates, and address missing/invalid records.
3. Metric calculation: compute margin, profit per unit, and contribution percentages.
4. Aggregation: group by product, division, and month where relevant.
5. Analysis: product ranking, division comparison, risk screening, and Pareto analysis.
6. Processed output: save validated calculations to CSV.
7. Dashboard: load processed data and render KPIs, charts, filters, and business answers.

Appendix B: Final Submission Checklist

- Verify total sales, gross profit, units, and transaction count against the final CSV.
- Confirm product names and division labels are standardized.
- Test that all filters update charts and business answers.
- Ensure chart titles, units, and labels are clear.
- Confirm State/Region analysis and the raw-data table are not included.
- Reconcile product and Pareto percentages with the dashboard output.